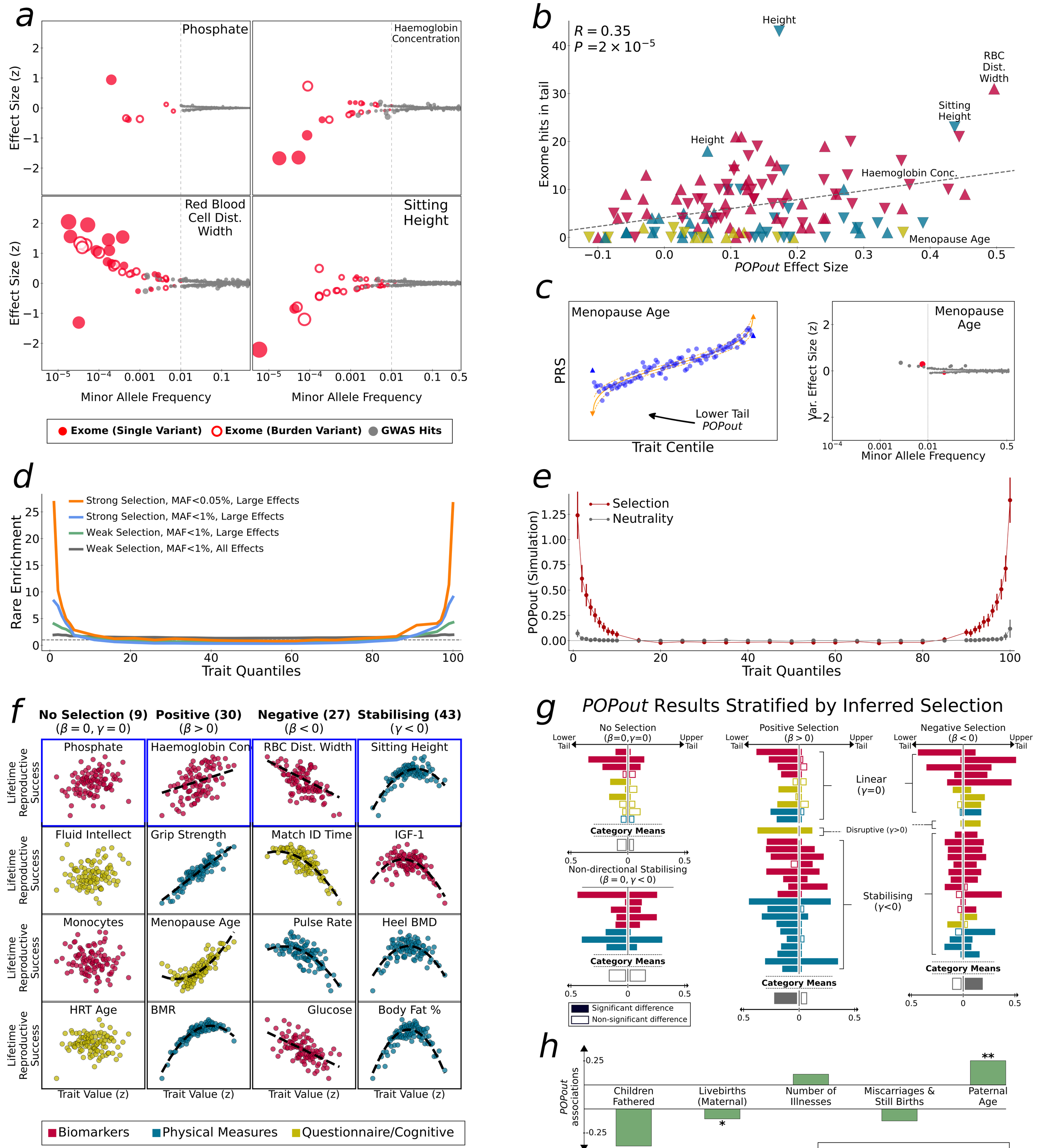
a
Phosphate
Haemoglobin Concentration
Red Blood Cell Dist. Width
Sitting Height
Effect Size (z)
Minor Allele Frequency
Exome (Single Variant)
Exome (Burden Variant)
GWAS Hits
b
R = 0.35
P = 2 × 10^−5
Height
RBC Dist. Width
Sitting Height
Height
Haemoglobin Conc.
Menopause Age
Exome hits in tail
POPout Effect Size
c
Menopause Age
PRS
Lower Tail POPout
Trait Centile
Menopause Age
Var. Effect Size (z)
Minor Allele Frequency
d
Strong Selection, MAF<0.05%, Large Effects
Strong Selection, MAF<1%, Large Effects
Weak Selection, MAF<1%, Large Effects
Weak Selection, MAF<1%, All Effects
Rare Enrichment
Trait Quantiles
e
Selection
Neutrality
POPout (Simulation)
Trait Quantiles
f
No Selection (9) (β = 0, γ = 0)
Positive (30) (β > 0)
Negative (27) (β < 0)
Stabilising (43) (γ < 0)
Lifetime Reproductive Success
Phosphate
Haemoglobin Con
RBC Dist. Width
Sitting Height
Fluid Intellect
Grip Strength
Match ID Time
IGF-1
Monocytes
Menopause Age
Pulse Rate
Heel BMD
HRT Age
BMR
Glucose
Body Fat %
Trait Value (z)
Biomarkers
Physical Measures
Questionnaire/Cognitive
g
POPout Results Stratified by Inferred Selection
No Selection (β=0,γ=0)
Positive Selection (β > 0)
Negative Selection (β < 0)
Lower Tail
Upper Tail
Linear (γ=0)
Disruptive (γ>0)
Stabilising (γ<0)
Category Means
Non-directional Stabilising (β = 0, γ < 0)
Significant difference
Non-significant difference
h
POPout associations
Children Fathered
Livebirths (Maternal)
Number of Illnesses
Miscarriages & Still Births
Paternal Age
*P < 0.05, **P < 0.01, ***P < 0.001